Artificial Intelligence Technology and Application

Machine Learning Lab Guide

Student Version

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1 Emotion Recognition of Customer Evaluations in the Retail Industry

1.1 Introduction

Emotion analysis is a classification technology based on natural language processing (NLP), and is usually used in classification methods for extracting emotional content of texts. Compared with related recommendation and precision marketing, users prefer to view or listen to the personal experience and feedback of users of the same type. For example, evaluations from users who have purchased similar products and comparison results from users who have used similar products can bring bidirectional values to users and enterprises. This experiment will discuss the problem and perform practice step by step from the perspectives of problem statement, breakdown, priority ranking, solution design, key point analysis, and summary and suggestions, and cultivate the project implementation thinking and implement analysis of the evaluation emotion analysis project from scratch.

1.1.1 Objectives

Upon completion of this task, you will be able to:

- Clarify the function and business value of emotion analysis.
- Understand the differences between conventional machine learning and deep learning in emotion analysis methods.
- Clarify label extraction methods for emotion analysis.
- Master deep learning-based emotion analysis methods.
- Understand future applications of emotion analysis.

1.1.2 Case background

The case in this document is for reference only. The actual procedure may vary. For details, see the corresponding product documents. Data engineer A works in the market data analysis department of a Hi-Tech company. The company plans to develop home appliance services, such as smart TV and smart reader, but it does not know how the current market is and how users evaluate such products. Therefore, the company wants the data department to output the market data survey report as soon as possible. Engineer A considers using the NLP technology to analyze users' evaluation tendency and evaluation keywords of competitors' products of the same type, and build an emotion prediction model to predict the users' emotion tendency based on texts.

1.2 Procedure

1.2.1 Data Management

The following information is involved:

- **Id**: ID
- **reviews.rating**: score
- **reviews.text**: text evaluation
- reviews.title: evaluation keywords
- **reviews.username**: name of the evaluator

This dataset contains 21 attribute fields and 34,657 data samples. The experiment aims to analyze customer evaluation data. Therefore, this document describes only the data attributes required in this experiment.

1.2.2 Data Reading

After obtaining the provided data files, you need to read and view the data over Python by performing the following steps:

Step 1 Import common library files such as **sklearn**, **pandas**, and **numpy**.

sklearn is a powerful third-party machine learning library of Python. It contains data in various aspects from data preprocessing to model training. Most functions in the **sklearn** library are classified into estimators and transformers. An estimator is equivalent to a modeling tool, and is used to predict data. Common estimator functions include **fit(x,y)** and **predict(x)**. A transformer is used to process data, such as reducing dimensions and standardizing data. Common transformer functions include **transform(x)** and **transform(x,y)**.

```
import pandas as pd
import matplotlib.pyplot as plt
import matplotlib as mpl
import nltk.classify.util
from sklearn.model_selection import train_test_split
from sklearn.metrics import confusion_matrix
from sklearn import metrics
from sklearn.metrics import roc_curve, auc
from nltk.classify import NaiveBayesClassifier
import numpy as np
import re
import string
import nltk
%matplotlib inline
```

Step 2 Read data from a local disk.

Step 3 Visualize the data.

Visualize the first five rows of data and view the data attribute columns.

Output:

	id	name	asins	brand	categories	keys	manufacturer	reviews.date	revie
0	AVqkIhwDv8e3D1O-lebb	All-New Fire HD 8 Tablet, 8 HD Display, Wi-Fi,...	B01AHB9CN2	Amazon	Electronics,iPad & Tablets,All Tablets,Fire Ta...	841667104676,amazon/53004484,amazon/b01ahb9cn2...	Amazon	2017-01-13T00:00:00.000Z	
1	AVqkIhwDv8e3D1O-lebb	All-New Fire HD 8 Tablet, 8 HD Display, Wi-Fi,...	B01AHB9CN2	Amazon	Electronics,iPad & Tablets,All Tablets,Fire Ta...	841667104676,amazon/53004484,amazon/b01ahb9cn2...	Amazon	2017-01-13T00:00:00.000Z	
2	AVqkIhwDv8e3D1O-lebb	All-New Fire HD 8 Tablet, 8 HD Display, Wi-Fi,...	B01AHB9CN2	Amazon	Electronics,iPad & Tablets,All Tablets,Fire Ta...	841667104676,amazon/53004484,amazon/b01ahb9cn2...	Amazon	2017-01-13T00:00:00.000Z	

This experiment uses only the **reviews.rating**, **reviews.text**, **reviews.username**, and **reviews.title** attribute columns. Therefore, you need to extract data from the dataset for the second time to retain only the attribute data in the four columns, and name the extracted attribute data file **permanent** to facilitate the subsequent experiment procedure.

View the missing values.

Output:

```
reviews.rating      33
reviews.text         1
reviews.title        5
reviews.username     2
dtype: int64
```

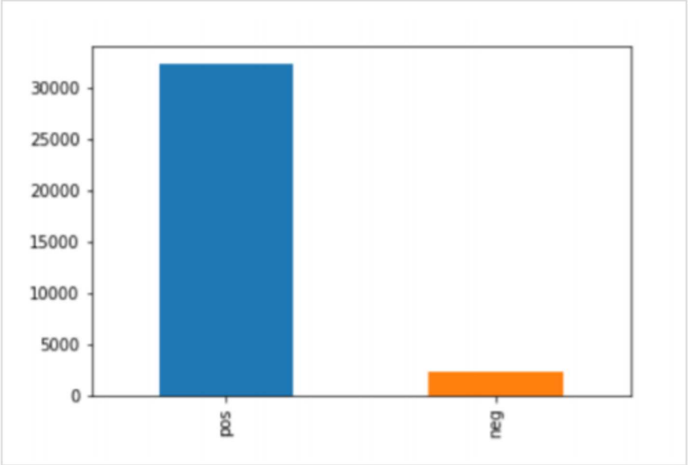
The **reviews.rating** attribute column is indispensable to emotion analysis. The dataset contains 34,657 data samples. The data volume is large. Therefore, you can delete the data samples with the **reviews.rating** value missing. Specifically, you can extract the data without the **reviews.rating** value and name the data **senti**, and extract the data with the **reviews.rating** value and name the data **check**.

With respect to score processing, this experiment defines data samples with the **reviews.rating** value greater than or equal to 4 as positive (**pos**) and those with the **reviews.rating** value less than 4 negative (**neg**), and renames the **reviews.rating** attribute column **senti**.

replace(x,y): replaces x with y.

Visualize the data after identifying the samples as positive or negative.

Output:



The output shows that the data is unbalanced.

1.2.3 Data Processing

A regular expression is used to check whether a string matches a pattern. The **re** module is added in Python 1.5 and later versions. The **re** module enables Python to have all regular expression functions:

- **re.sub()**: match item in the string to be replaced.

  re.sub(pattern, repl, string, count=0, flags=0)
- **pattern**: pattern string in the regular expression.
- **repl**: character string to be replaced, which can also be a function.
- **string**: original character string to be searched for and replaced.
- **count**: maximum number of replacements after pattern matching. The default value is **0**, indicating that all matches need to be replaced.

Data slicing is to use **pandas.DataFrame.sample** to randomly select several rows of data.

```
```

- **n**: number of rows to be extracted.
- **frac**: proportion of rows to be extracted. If **frac** is set to **0.8**, 80% of the rows need to be extracted.
- **replace**: indicates whether extraction is performed with replacement. The value **True** indicates that extraction is performed with replacement.
- **random_state**: seed of the random number generator. If **random_state** is set to **None**, the obtained data is not repeated.

Step 1 Import related packages.

```
import nltk.classify.util
from nltk.classify import NaiveBayesClassifier
import numpy as np
import re
import string
import nltk
```

Step 2 Preprocess data.

Text data includes spaces, punctuation marks, and data. This experiment focuses on the text (English) analysis. Therefore, you need to delete the information other than letters. You can define the **cleanup()** function, use a regular expression to delete non-letter characters, use the **lower()** function to convert uppercase letters into lowercase ones, and delete spaces, including **'\n'**, **'\r'**, **'\t'**, and **' '**. After **apply()** is used, the **reviews.text** attribute is saved as the **summary_clean** column.

```
```

Step 3 Generate a training set and a test set.

Obtain **["Summary_Clean","senti"]** from the **senti** dataset and save it as the **split** dataset.

```
```

Output:

```
In [17]: split
```

Out[17]:

	Summary_Clean	senti
0	this product so far has not disappointed my ch...	pos
1	great for beginner or experienced person bough...	pos
2	inexpensive tablet for him to use and learn on...	pos
3	i ve had my fire hd two weeks now and i love i...	pos
4	i bought this for my grand daughter when she c...	pos
5	this amazon fire inch tablet is the perfect si...	pos
6	great for e reading on the go nice and light w...	pos
7	i gave this as a christmas gift to my inlaws h...	pos
8	great as a device to read books i like that it...	pos

Use 80% of data in **split** as the training set through **split.sample()**, remove the data that has been used in the training set **train** from **split** through **drop()**, and use the remaining data as the test set **test**.

Output:

Size of train:

```
In [23]: train.count()

Out[23]: Summary_Clean    27702
         senti            27702
         dtype: int64
```

Size of test:

```
In [25]: test.count()

Out[25]: Summary_Clean    6925
         senti            6925
         dtype: int64
```

1.2.4 Model Training

1.2.4.1 Model Selection

Emotion analysis of customer evaluations is essentially a classification problem, which can be solved by using a classification model. Practice has proved that a Naive Bayes model based on all words performs well in solving some problems, while a model using a word subset performs well in solving other problems. Logistic regression (LR), multinomial NB, and Bernouli NB are selected based on comprehensive consideration.

1.2.4.2 Model Calculation and Evaluation

The Naive Bayes classifier usually uses three models: Gaussian model, Polynomial model, and Bernoulli model. The three models respectively correspond to functions **GaussianNB()**, **MultinomialNB()**, and **BernoulliNB()** in **sklearn**.

```
from sklearn.naive_bayes import GaussianNB

from sklearn.naive_bayes import MultinomialNB
```

- **GaussianNB()** is used when a feature is a continuous variable.
- **MultinomialNB()** is used when a feature is a discrete variable.
- **BernoulliNB()** is used when a feature is a discrete variable and the feature can be set only to **1** or **0**.

The model calculation phase consists of the following steps:

Step 1 Import libraries.

```
from wordcloud import STOPWORDS
from sklearn.feature_extraction.text import TfidfTransformer
from sklearn.feature_extraction.text import CountVectorizer
```

Step 2 Build and evaluate a model.

Convert the data in the training set, test set, and verification set into a list and create indexes.

```

```

Set all words in **train["words"]** to **True** and add **neg** or **pos** to the end of a sentence based on the scoring criteria.

```

```

```
deleted = True          neg : pos    =     51.3 : 1.0
warning = True          neg : pos    =     51.3 : 1.0
   bent = True          neg : pos    =     42.0 : 1.0
   nope = True          neg : pos    =     42.0 : 1.0
 rotate = True          neg : pos    =     42.0 : 1.0
```

Use a trained classifier to attach emotion labels to the test set and verification set to predict whether words in the test set and verification set are positive or negative.

Output:

```
In [73]: prediction["Naive"]

Out[73]: array(['neg', 'neg', 'neg', ..., 'neg', 'neg', 'neg'], dtype='<U3')
```

Output:

```
In [74]: check["Naive"]

Out[74]: 0     neg
         1     neg
         2     neg
         3     neg
         4     neg
         5     neg
         6     neg
         7     neg
         8     pos
         9     neg
         10    neg
         11    pos
         12    neg
         13    neg
         14    neg
         15    neg
         16    pos
```

The original dataset **check** does not contain **review.ratings** data. As shown in the preceding figure, whether each word is negative or positive is predicted after the classifier is created based on the training set.

```
from sklearn.naive_bayes import MultinomialNB
```

Use the CountVectorizer class to perform vectorization, invoke the TfidfTransformer class to perform preprocessing, construct the term frequency (TF) vector, and calculate the importance of words. The training set, test set, and verification set are obtained, which are **X_train_tfidf**, **X_test_tfidf**, and **checktfidf**, respectively.

The main idea of TF is as follows: If a word or phrase has a high TF in an article but a low TF in other retail articles, the word or phrase is considered to have a good class distinguishing capability. TF-IDF tends to filter out commonly used words and retain important words.

The CountVectorizer class converts words in the text into a TF matrix, and uses the **fit_transform()** function to calculate the number of appearance times of each word. In general, you can use CountVectorizer to extract features and then use TfidfTransformer to calculate the weight of each feature.

Retain important words based on **tfidf_transformer** to construct the MultinomialNB model.

Output:

```
Multinomial Accuracy : 0.9329963898916968
```

Retain important words based on **tfidf_transformer** to construct the BernoulliNB model.

```
from sklearn.naive_bayes import BernoulliNB
```

Output:

```
Bernoulli Accuracy : 0.9204332129963899
```

Retain important words based on **tfidf_transformer** to construct the LR model.

```
from sklearn import linear_model
```

Output:

```
Logistic Regression Accuracy : 0.9377617328519856
```

In comparison, the LR model has higher accuracy than the other two models.

Step 3 Verify the model.

Select the LR model for verification.

Output:

```
Sample estimated as POS: negative prob 0.000000, positive prob 1.000000
Sample estimated as NEG: negative prob 0.993617, positive prob 0.006383
Sample estimated as NEG: negative prob 0.964536, positive prob 0.035464
```

The classifier accurately provides the positive probability and negative probability of each sentence.

Step 4 Build a word cloud.

```
from wordcloud import WordCloud, STOPWORDS
```

Output:

----End